

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                             |                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Product Name</b>         | <b><u>Tetrahydrofuran, Chromplete™ for HPLC, ACS and Spectrophotometry</u></b>                                             |
| <b>CAS No</b>               | 109-99-9                                                                                                                   |
| <b>Synonyms</b>             | THF                                                                                                                        |
| <b>Molecular Formula</b>    | C <sub>4</sub> H <sub>8</sub> O                                                                                            |
| <b>Molecular Weight</b>     | 72.11                                                                                                                      |
| <b>Recommended Use</b>      | Laboratory chemicals.                                                                                                      |
| <b>Uses advised against</b> | SU21 - Consumer uses: Private households (= general public = consumers) REACH Annex XVII Restriction - refer to SECTION 15 |

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>T001092500; T001091000; T001094000</b>                                                   |
| <b>Address</b>                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| <b>Telephone / Fax Numbers</b> | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| <b>E-mail address</b>          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

**HSNO Approval Number**     **HSR001224**

### GHS Classification

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Acute Oral Toxicity

Category 4

Skin Corrosion/Irritation

Category 2

Serious Eye Damage/Eye Irritation

Category 2

Carcinogenicity

Category 2

Specific target organ toxicity - (single exposure)

Category 3

#### Environmental hazards

Based on available data, the classification criteria are not met

**Label Elements**



**Signal Word**

**Danger**

**Hazard Statements**

H225 - Highly flammable liquid and vapor  
H302 - Harmful if swallowed  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H336 - May cause drowsiness or dizziness  
H351 - Suspected of causing cancer  
H315 - Causes skin irritation

**Precautionary Statements**

**Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear eye protection/ face protection

**Response**

P301 + P312 - IF SWALLOWED: Call a POISON CENTER or doctor if you feel unwell  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors  
May form explosive peroxides

## Section 3 - Composition and Information on Ingredients

| Component       | CAS No   | Weight % |
|-----------------|----------|----------|
| Tetrahydrofuran | 109-99-9 | >95      |

## Section 4 - First Aid Measures

### Description of first aid measures

|                                            |                                                                                                                                                        |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                 |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                             |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                           |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                        |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                           |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.       |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                   |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Causes central nervous system depression |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                        |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. May form explosive peroxides.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), peroxides.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

#### Environmental Precautions

Should not be released into the environment.

#### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

#### Precautions to prevent secondary hazards

Clean contaminated objects and areas thoroughly observing environmental regulations

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

### Precautions for Safe Handling

#### Advice on safe handling

Not suitable for concentration or distillation. May form explosive peroxides on prolonged storage. If peroxide formation is suspected, do not open or move container. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### Conditions for Safe Storage, Including any Incompatibilities

#### Storage Conditions

Store under an inert atmosphere. Shelf life 12 months (Unopened) or Shelf life: 3 months after opening. Containers should be dated when opened. May form explosive peroxides on prolonged storage. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

#### Incompatible Materials

Strong oxidizing agents. Acids.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

### Control parameters

#### Exposure limits

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component       | New Zealand WEL                                                                                   | Australia                                  | ACGIH TLV                            | The United Kingdom                                                                                                        |
|-----------------|---------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup><br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>Skin | TWA: 100 ppm<br>TWA: 295 mg/m <sup>3</sup> | TWA: 50 ppm<br>STEL: 100 ppm<br>Skin | STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 150 mg/m <sup>3</sup> 8 hr<br>Skin |

### Biological limit values

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for Chemical Substances and Physical Agents & Biological Exposure Indices. 2022 Edition

| Component       | New Zealand                                                                                | Australia | ACGIH - Biological Exposure Indices                                           | United Kingdom |
|-----------------|--------------------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------|----------------|
| Tetrahydrofuran | 2 mg/g creatinine (urine) end of exposure or shift, within 1 hour of end of exposure (THF) |           | 2 mg/L<br>Medium: urine<br>Time: end of shift<br>Determinant: Tetrahydrofuran |                |

### Appropriate engineering controls

#### Engineering Measures

Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Individual protection measures, such as personal protective equipment

**Eye Protection** Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material  | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                                                             |
|-----------------|-------------------|-----------------|-----------------|----------------------------------------------------------------------------------------------------------------------------|
| Butyl rubber.   | < 25 minutes      | 0.6 mm          | AS/NZS 2161     | Permeation rate 106 µg/cm <sup>2</sup> /min As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Neoprene gloves | < 15 minutes      | 0.45 mm         |                 |                                                                                                                            |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                                               |                                          |
|------------------------------------------------|-----------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                                        |                                          |
| <b>Appearance</b>                              | Colorless                                     |                                          |
| <b>Odor</b>                                    | Petroleum distillates                         |                                          |
| <b>Odor Threshold</b>                          | No data available                             |                                          |
| <b>pH</b>                                      | 7-8                                           | 20% aq. solution                         |
| <b>Melting Point/Range</b>                     | -108.4 °C / -163.1 °F                         |                                          |
| <b>Softening Point</b>                         | No data available                             |                                          |
| <b>Boiling Point/Range</b>                     | 66 °C / 150.8 °F                              |                                          |
| <b>Flammability (liquid)</b>                   | Highly flammable                              | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                                | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.5 vol%<br><b>Upper</b> 12 vol% |                                          |
| <b>Flash Point</b>                             | -21 °C / -5.8 °F                              | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 215 °C / 419 °F                               |                                          |
| <b>Decomposition Temperature</b>               | No data available                             |                                          |
| <b>Viscosity</b>                               | 0.456 mPas @ 20°C Dynamic                     |                                          |
| <b>Water Solubility</b>                        | Miscible                                      |                                          |
| <b>Solubility in other solvents</b>            | No information available                      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                               |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                |                                          |
| Tetrahydrofuran                                | 0.45                                          |                                          |
| <b>Vapor Pressure</b>                          | 170 mbar @ 20 °C                              |                                          |
| <b>Density / Specific Gravity</b>              | 0.880                                         |                                          |
| <b>Bulk Density</b>                            | Not applicable                                | Liquid                                   |
| <b>Vapor Density</b>                           | 2.5 (Ether = 1.0)                             | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)                       |                                          |
| <b><u>Other information</u></b>                |                                               |                                          |
| <b>Molecular Formula</b>                       | C4 H8 O                                       |                                          |
| <b>Molecular Weight</b>                        | 72.11                                         |                                          |
| <b>Explosive Properties</b>                    | Vapors may form explosive mixtures with air   |                                          |
| <b>Evaporation Rate</b>                        | > 1 (Ether = 1.0) - (Butyl Acetate = 1.0)     |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                                                                 |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | Yes. May form explosive peroxides                                                                                                               |
| <b>Stability</b>                        | Stable under recommended storage conditions. Reacts with air to form peroxides. May form explosive peroxides on prolonged storage. Hygroscopic. |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                        |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                        |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization may occur.                                                                                                             |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                                                                   |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition, Exposure to moist air or water.           |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Acids.                                                                                                                 |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). peroxides.                                                                             |

## Section 11 - Toxicological Information

### Acute Effects

**Information on likely routes of exposure****Product Information**

|                   |                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Irritating to respiratory system. May be harmful if inhaled. INHALATION MAY CAUSE CENTRAL NERVOUS SYSTEM EFFECTS. |
| <b>Eyes</b>       | Irritating to eyes.                                                                                               |
| <b>Skin</b>       | May be harmful in contact with skin. May cause irritation.                                                        |
| <b>Ingestion</b>  | Harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea.             |

**Numerical measures of toxicity****(a) acute toxicity;**

|                   |                                                                  |
|-------------------|------------------------------------------------------------------|
| <b>Oral</b>       | Category 4                                                       |
| <b>Dermal</b>     | Based on available data, the classification criteria are not met |
| <b>Inhalation</b> | Based on available data, the classification criteria are not met |

| Component       | LD50 Oral          | LD50 Dermal           | LC50 Inhalation                               |
|-----------------|--------------------|-----------------------|-----------------------------------------------|
| Tetrahydrofuran | 1650 mg/kg ( Rat ) | > 2000 mg/kg (Rabbit) | 180 mg/L ( Rat ) 1 h<br>53.9 mg/L ( Rat ) 4 h |

**(b) skin corrosion/irritation;** Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;** Category 2

**(d) respiratory or skin sensitization;**

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>Respiratory</b> | Based on available data, the classification criteria are not met |
| <b>Skin</b>        | Based on available data, the classification criteria are not met |

| Component                           | Test method                                       | Test species | Study result    |
|-------------------------------------|---------------------------------------------------|--------------|-----------------|
| Tetrahydrofuran<br>109-99-9 ( >95 ) | Local Lymph Node Assay OECD<br>Test Guideline 429 | mouse        | non-sensitising |

**(e) germ cell mutagenicity;** Based on available data, the classification criteria are not met

| Component                           | Test method                                             | Test species          | Study result |
|-------------------------------------|---------------------------------------------------------|-----------------------|--------------|
| Tetrahydrofuran<br>109-99-9 ( >95 ) | OECD Test Guideline 476<br>Gene cell mutation           | in vivo<br>Mammalian  | negative     |
|                                     | OECD Test Guideline 473<br>Chromosomal aberration assay | in vitro<br>Mammalian | negative     |

**(f) carcinogenicity;** Category 2  
Limited evidence of a carcinogenic effect

| Component       | New Zealand             | Australia | New South<br>Wales | Western<br>Australia | IARC     | EU | UK | Germany |
|-----------------|-------------------------|-----------|--------------------|----------------------|----------|----|----|---------|
| Tetrahydrofuran | Suspected<br>carcinogen |           |                    |                      | Group 2B |    |    |         |

**(g) reproductive toxicity;** Based on available data, the classification criteria are not met

| Component                           | Test method             | Test species / Duration | Study result      |
|-------------------------------------|-------------------------|-------------------------|-------------------|
| Tetrahydrofuran<br>109-99-9 ( >95 ) | OECD Test Guideline 416 | Rat 2 Generation        | NOAEL = 3,000 ppm |

**(h) STOT-single exposure;** Category 3  
**Results / Target organs** Respiratory system

Central nervous system (CNS)

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed**

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Causes central nervous system depression.

## Section 12 - Ecological Information

### Ecotoxicity

Aquatic ecotoxicity Do not empty into drains. .

| Component       | Freshwater Fish                                                                        | Water Flea                                   | Freshwater Algae | Microtox |
|-----------------|----------------------------------------------------------------------------------------|----------------------------------------------|------------------|----------|
| Tetrahydrofuran | 2160 mg/l LC50 = 96 h<br>Pimephales promelas<br>Leuciscus idus: LC50:<br>2820 mg/L/48h | EC50 48 h 3485 mg/l<br>EC50: >10000 mg/L/24h |                  |          |

Terrestrial ecotoxicity There is no data for this product

Persistence and Degradability Product is biodegradable

Persistence Persistence is unlikely, based on information available.

Degradation in sewage treatment plant Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

Bioaccumulative Potential Bioaccumulation is unlikely

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Tetrahydrofuran | 0.45    | No data available             |

Mobility The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air

### Other adverse effects

### Endocrine Disruptor Information

| Component       | EU - Endocrine Disruptors<br>Candidate List | EU - Endocrine Disruptors -<br>Evaluated Substances | Japan - Endocrine Disruptor<br>Information |
|-----------------|---------------------------------------------|-----------------------------------------------------|--------------------------------------------|
| Tetrahydrofuran | Group III Chemical                          |                                                     |                                            |

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste treatment methods

Waste from Residues/Unused Do not allow into drains or watercourses or dispose of where ground or surface waters may



|                               |                                                                                                                                                                                                                                                                                                                               |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Products</b>               | be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                                                                    |
| <b>Contaminated Packaging</b> | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                      |
| <b>Other Information</b>      | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. |

## Section 14 - Transport Information

| Component                           | Hazchem Code |
|-------------------------------------|--------------|
| Tetrahydrofuran<br>109-99-9 ( >95 ) | 2YE          |

### NZS 5433:2020

|                             |                 |
|-----------------------------|-----------------|
| <b>UN-No</b>                | UN2056          |
| <b>Proper Shipping Name</b> | TETRAHYDROFURAN |
| <b>Hazard Class</b>         | 3               |
| <b>Packing Group</b>        | II              |

### IATA

|                             |                 |
|-----------------------------|-----------------|
| <b>UN-No</b>                | UN2056          |
| <b>Proper Shipping Name</b> | TETRAHYDROFURAN |
| <b>Hazard Class</b>         | 3               |
| <b>Packing Group</b>        | II              |

### IMDG/IMO

|                             |                 |
|-----------------------------|-----------------|
| <b>UN-No</b>                | UN2056          |
| <b>Proper Shipping Name</b> | TETRAHYDROFURAN |
| <b>Hazard Class</b>         | 3               |
| <b>Packing Group</b>        | II              |

**Environmental hazards** No hazards identified

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code** Not applicable, packaged goods

**Special Precautions** No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

**Additional information** None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                             |           |
|-----------------------------|-----------|
| <b>HSNO Approval Number</b> | HSR001224 |
|-----------------------------|-----------|

### **National Regulations**

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

**Certified handlers, tracking and controlled substance license requirements**

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

| Component       | New Zealand          |
|-----------------|----------------------|
| Tetrahydrofuran | Suspected carcinogen |

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Authorisation/Restrictions  
according to EU REACH**

| Component       | REACH (1907/2006) - Annex XIV -<br>Substances Subject to<br>Authorization | REACH (1907/2006) - Annex XVII -<br>Restrictions on Certain Dangerous<br>Substances | REACH Regulation (EC<br>1907/2006) article 59 - Candidate<br>List of Substances of Very High<br>Concern (SVHC) |
|-----------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | -                                                                         | Use restricted. See item 75.<br>(see link for restriction details)                  | -                                                                                                              |

<https://echa.europa.eu/substances-restricted-under-reach>

**International Inventories**

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component       | CAS No   | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-----------------|----------|-------|------|-----------|--------|-----|----------|-------|------|
| Tetrahydrofuran | 109-99-9 | X     | X    | 203-726-8 | -      | -   | KE-33454 | X     | X    |

| Component       | CAS No   | TSCA | TSCA Inventory<br>notification -<br>Active-Inactive | DSL | NDL | PICCS | ISHL | ENCS |
|-----------------|----------|------|-----------------------------------------------------|-----|-----|-------|------|------|
| Tetrahydrofuran | 109-99-9 | X    | ACTIVE                                              | X   | -   | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

**Legend**

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**AICS** - Australian Inventory of Chemical Substances  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**NZS 5433:2020** - Transport of Dangerous Goods on Land  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**PNEC** - Predicted No Effect Concentration  
**OECD** - Organisation for Economic Co-operation and Development  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**ADG** - Australian Code for the Transport of Dangerous Goods by Road and Rail  
**LC50** - Lethal Concentration 50%  
**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).  
<https://echa.europa.eu/information-on-chemicals>  
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS  
EPA Guide to classifying hazardous substances in New Zealand  
EPA - Assigning a product to an existing HSNO approval guide

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
First aid for chemical exposure, including the use of eye wash and safety showers.  
Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.  
Chemical incident response training.

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#### Disclaimer

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**End of Safety Data Sheet**